ON THE EFFECTIVENESS OF SELF-SUPERVISED MUSIC REPRESENTATIONS AND DEEP METRIC LEARNING FOR COVER SONG RETRIEVAL*

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ABSTRACT

This report presents a systematic study on adapting self-supervised music models (MERT) for Cover Song Identification (CSI) using deep metric learning. To mitigate the domain gap caused by timbral variations between originals and covers, we introduce a trainable projection head trained on a group-aware split with triplet and contrastive losses. We perform exhaustive ablation studies on temporal segmentation, temporal pooling, layer selection, loss formulations, and regularizations (BatchNorm and feature augmentations). Furthermore, we evaluate dynamic sequence alignment (DTW) and learned cross-similarity classifiers, discussing the critical role of representation constraints and the impact of extreme compression via chroma bottlenecks.

Keywords Cover Song Identification · Self-Supervised Learning · Deep Metric Learning · MERT · Dynamic Time Warping

1 Introduction

Cover Song Identification (CSI) is a complex challenge in audio retrieval because it requires a model to bridge a severe acoustic domain gap. A cover song retains the abstract musical semantics of the original track (such as melody, harmony, and structural progression) but often introduces radical changes to the surface-level acoustics, including entirely different instrumentation, tempo, and vocal styles. This extreme acoustic variance causes standard audio feature extractors to project the original and the cover into completely disparate regions of the latent embedding space.

Historically, the MIR community addressed this gap by relying on handcrafted features, such as Chromagrams, which explicitly collapse timbral information to isolate harmonic progressions. While robust, these methods are rigid and require computationally expensive sequence-alignment algorithms like Dynamic Time Warping (DTW) for retrieval. Recently, large-scale self-supervised learning (SSL) models like MERT have set new benchmarks in audio understanding by learning rich, hierarchical representations directly from raw waveforms. However, because SSL models are typically pre-trained to reconstruct acoustic features, they perform poorly on CSI out-of-the-box. Without explicit guidance, MERT interprets the timbral differences between covers as fundamental dissimilarities.

To overcome this limitation, our methodology leverages Deep Metric Learning to semantically align covers within the embedding space. Rather than fine-tuning the entire multi-million parameter backbone, we freeze the pre-trained MERT encoder and introduce a custom, trainable projection head. By optimizing this projection head with triplet and contrastive loss functions, we force the network to discard surface-level timbral variance and map different versions of the same song close together.

In this work, we present a complete pipeline for self-supervised cover song retrieval. We constructed a custom, group-aware dataset of 120 track pairs explicitly curated to challenge the model with diverse instrumental and vocal variations. To prevent dimensional collapse in the embedding space, we introduce a suite of temporal segmentation strategies

*Code repository: https://github.com/TheodoraPav/CoverSongIdentification and apply augmentations at both the waveform and feature levels. Finally, we conduct exhaustive ablation studies comparing standard Triplet, Triplet Hard, and InfoNCE loss formulations to identify the most robust configuration for mapping musical identity.

2 Related Work

2.1 Traditional Cover Song Identification

Early approaches to cover song identification focused on extracting features that are invariant to timbre and instrumentation. The most common representations are Chromagrams and Harmonic Pitch Class Profiles (HPCP) [2], which map audio signals into 12-dimensional pitch-class bins. These features isolate the harmonic progression of a song. To compare two songs, researchers typically relied on sequence alignment algorithms, with Dynamic Time Warping (DTW) [6] being the standard method to handle tempo variations. While these handcrafted methods are highly interpretable, they often fail when covers introduce significant structural changes or complex background noise.

2.2 Self-Supervised Music Representations

In recent years, self-supervised learning has become the standard for extracting generic audio features. Models are pre-trained on large amounts of unlabeled audio to reconstruct masked acoustic features. For this project, we utilize MERT [1], a leading self-supervised acoustic model. MERT is unique because it is trained using pseudo-labels generated from both acoustic models (like RVQ-VAE) and musical models (like Constant-Q Transform). This dual-objective training allows the model's transformer layers to capture both low-level acoustic details and high-level musical semantics, making it a strong candidate for music retrieval tasks.

2.3 Deep Metric Learning for Audio Retrieval

Deep metric learning focuses on mapping data points to an embedding space where similar items are close together and dissimilar items are far apart. This is typically achieved using contrastive or triplet-based loss functions. In the context of audio retrieval, methods like NT-Xent (InfoNCE) are widely used to pull augmented versions of the same audio clip together. Recent cover song identification systems, such as ByteCover [3] and CoverHunter [4], have successfully applied these techniques by treating different versions of a song as positive pairs. Our work builds on this by explicitly comparing multiple loss formulations, including standard Triplet, Triplet Hard, and Proxy-Anchor loss, to determine the most stable objective for cover song retrieval.

3 Proposed Methodology

An overview of our complete cover song identification pipeline is illustrated in Figure 1. The system processes raw audio into hierarchical embeddings via a frozen MERT backbone, pools the temporal representations, projects them into a contrastive latent space using a custom MLP (where we explicitly evaluate dimensionality bottlenecks), and finally aligns them using track-level matching strategies.

Figure 1: Overview of our proposed Cover Song Identification pipeline. The frozen MERT backbone extracts multi-layer hierarchical features which are aggregated by our custom LayerPooler. A projection head reduces the dimensionality into a contrastive latent space, optimized using our Modified Triplet Hard loss alongside waveform and feature-level augmentations.

3.1 Temporal Segmentation and Preprocessing

To construct valid anchor, positive, and negative pairs from audio tracks, we propose a dynamic temporal segmentation strategy. Each audio track is initially converted to a mono signal at 24 kHz (matching the MERT preprocessor). Because cover songs exhibit significant structural and tempo variations (e.g., an extended acoustic intro or a tempo shift), extracting rigid fixed-length sequences can lead to severely misaligned comparisons.

To address this, we evaluate four segmentation techniques:

- **Random:** Uniformly random T-second segments are extracted from anywhere in the track.
- **Stratified:** The track is divided into N equal-duration temporal zones, and one segment is randomly sampled from within each zone. This ensures that the entire structural progression of the song is represented in the dataset.
- **Mixed:** Segments of variable length, uniformly drawn between T and $2T$ seconds, are extracted. This variation forces the model to learn duration-invariant embeddings.
- **Beat-Aligned:** To explicitly counter tempo variations between covers, we employ the `librosa` library [7] to detect the tempo grid and snap segment boundaries to the nearest beat, ensuring segments encompass complete musical phrases rather than arbitrary acoustic frames.

Finally, independently of the segmentation *strategy*, we compare two ways of exposing segments to the trainer each epoch (ablated as *Dynamic Pool Feeding* in Table 2). This concerns **which temporal crops are active during training**, and is distinct from frame-level mean/max pooling (Section 3.2) and from contrastive mini-batch construction (Section 3.7):

- **Fixed Segment Pool:** A fixed set of N segments per track is preprocessed and cached once. Every training epoch uses exactly this same set, no resampling across epochs. This is the simplest setup but encourages memorization of specific temporal crops.
- **Dynamic Segment Pool:** We preprocess a larger duration-capped candidate pool per track (up to 40 segments). Loading the full pool every epoch is infeasible on GPU memory, so at the start of each training epoch we randomly draw a smaller active subset (N=10 segments per track). Validation keeps a fixed, deterministic subset of the same size for reproducible metrics. Resampling different zone combinations across epochs acts as temporal data augmentation and reduces overfitting to fixed crops.

Our final pipeline adopts dynamic 10/40 feeding; Table 2 compares this against a fixed 10-segment pool once backbone, sampling, pooling, and loss settings have been selected by earlier ablations.

3.2 Temporal Feature Pooling

Because the transformer backbone outputs a sequence of frame-level embeddings for each audio segment, we must collapse the temporal dimension to obtain a single descriptive vector before performing metric learning. We explicitly evaluate two temporal pooling strategies as an initial architectural ablation:

- **Mean Pooling:** Averaging all frames across the segment to capture the overall musical context and continuous harmonic flow.
- **Max Pooling:** Extracting the maximum activation across the temporal dimension to isolate the most important, distinctive acoustic features.

3.3 Backbone Layer Aggregation and Softmax Blend

Each audio segment is encoded by a frozen MERT-95M backbone [1]; we do not fine-tune its 95M parameters during CSI training. Only the projection head and *LayerPooler* receive gradient updates, isolating the contribution of deep metric learning from backbone pre-training. Self-supervised transformer architectures like MERT encode highly hierarchical information. As demonstrated by the original MERT authors and further explored by the MERIT framework [8], different transformer layers specialize in different levels of abstraction: lower layers primarily capture raw acoustic and timbral features, while middle and upper layers capture more abstract musical semantics, such as pitch, rhythm, and melody. Building upon the findings of MERIT (which explicitly disentangles these representations) we recognize that cover song identification fundamentally requires isolating melodic and rhythmic semantics while actively ignoring timbral noise. Therefore, the choice of which MERT layer outputs to utilize is critical.

To empirically evaluate this hypothesis for cover song retrieval, we implement a custom *LayerPooler* module. Rather than relying solely on the final transformer block, we extract the temporally-pooled outputs from all 13 layers of the MERT backbone (including the initial embedding layer). We then group these layers into predefined subsets corresponding to different theoretical levels of musical abstraction:

- **Acoustic:** Layers 1 to 3, expected to capture low-level timbre.
- **Early-Mid:** Layers 4 to 6.

- **Mid-Beat / Mid-Pitch:** Layers 7 to 9, expected to capture rhythmic and harmonic structures.
- **Semantic:** Layers 10 to 12, expected to capture high-level musical semantics.
- **Musical Core:** Layers 4 to 9, a broader subset targeting the middle of the network.
- **Mean-All:** A uniform average of all 13 layers.

In addition to these explicit subsets, we implement a **Learned Mix** strategy, which introduces a trainable softmax parameter over the 13 layers. This allows the network to dynamically learn the optimal layer weighting during the metric learning optimization process, serving as a robust baseline against our handcrafted layer groupings. Because these softmax parameters govern the fundamental feature distribution across all layers, they are highly sensitive to gradient starvation. To address this, we assign the *LayerPooler* to a distinct parameter group with an elevated learning rate during optimization, ensuring the network actively explores different layer combinations rather than freezing at the initial uniform distribution.

3.4 Projection Head and Representation Bottlenecks

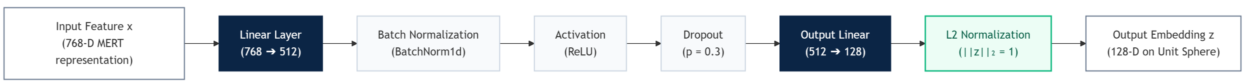


Figure 2: Architecture of the custom MLP projection head. The 768-dimensional MERT features are passed through a bottleneck layer, batch normalization, ReLU activation, and dropout ($p = 0.3$), before being projected to the final 128-dimensional hypersphere.

To adapt the frozen MERT representations for contrastive learning, we append a custom Multilayer Perceptron (MLP) projection head. As detailed in Figure 2, the pooled outputs from the transformer layers ($d = 768$) are first passed through a fully connected layer reducing the dimension to a hidden layer of 512, followed by an optional 1D Batch Normalization layer, ReLU activation, and Dropout ($p = 0.3$). A final linear transformation projects the features into the desired 128-dimensional embedding space before applying L_2 normalization, ensuring that all embeddings reside on the unit hypersphere for cosine similarity comparisons. Whether BN is enabled is an ablation switch documented in Section 3.7.

3.5 Information Bottlenecks and Chroma Capacity

Crucially, the dimensionality of the final projection space serves as an *Information Bottleneck*. We hypothesize that projecting the high-dimensional acoustic features into a severely restricted space (e.g., $d = 24$, $d = 48$, or $d = 96$) forces the network to discard complex timbral and instrumental noise, preserving only the most vital structural components of the audio. For example, compressing the representation down to 24 dimensions conceptually mirrors a traditional Chromagram (which spans two octaves of 12 semitones). By employing this bottleneck, we actively guide the network to construct a purely harmonic and melodic latent space, which is essential for bridging the domain gap between cover songs.

3.6 Data Augmentation and Regularization

Deep Metric Learning on audio embeddings is highly susceptible to dimensional collapse, where the network projects all inputs to a single point to trivially solve the contrastive objective. To prevent this, we introduce augmentation strategies at both the waveform and feature levels.

Waveform-Level Augmentation: Prior to feature extraction, we apply offline transformations to the raw audio signals using the `audiomentations` library. This includes *PitchShift* (to simulate different vocal ranges or keys), *TimeStretch* (to simulate different tempos), and *AddGaussianSNR* (to simulate recording noise). By applying these transformations, we actively teach the model that pitch and tempo variations do not alter the underlying musical identity.

Feature-Level Augmentation: Because extracting MERT representations online is computationally prohibitive, we introduce augmentation directly within the projection head: Gaussian noise and Dropout ($p = 0.3$) on the intermediate 768-dimensional embeddings before the final L_2 normalization.

3.7 Additional Ablations: Mini-Batch Construction and Batch Normalization

Beyond the core embedding pipeline, we run focused ablations on *how* contrastive mini-batches are built and on an optional normalization layer in the projection head. Results appear in Tables 4–5. These training-time choices are independent of the group-aware train/validation split (Section 4.1), which only controls which song groups appear in each split.

GroupPairBatchSampler. With the default PyTorch shuffle loader, segments are drawn uniformly at random. Triplet and contrastive losses then receive no guarantee that an anchor's paired cover appears in the same mini-batch, so many updates lack usable positives. Our *GroupPairBatchSampler* constructs each batch from P song groups: for every group, it samples K segments from the original track and K from its paired cover ($2PK$ segments total). Negatives for a given anchor come from the other groups in the batch. The pair (P, K) is resolved automatically from the training batch size and segment budget. We ablate this sampler against shuffle batching in Table 4.

Batch Normalization (BN). When enabled, a 1D Batch Normalization layer sits in the projection MLP after the first linear map (Figure 2). BN normalizes hidden activations using the mean and variance of the current mini-batch, with learnable affine parameters. In metric learning, this can stabilize gradients and reduce embedding collapse at the cost of a small ranking drop. We treat BN as an on/off ablation across loss functions (Table 3) and as a sequential pipeline component (Table 4).

3.8 Deep Metric Learning Loss Formulations

To fulfill the objective of mapping different versions of the same song close together in the embedding space, we evaluate four primary metric learning objectives. Let $f_\theta(\cdot)$ denote the frozen MERT backbone combined with our trainable projection head, mapping an input segment x to a d-dimensional L_2-normalized embedding $z \in \mathbb{R}^d$. Mini-batches follow either random shuffle or the *GroupPairBatchSampler* (Section 3.7). Let an anchor segment be x_a, a positive cover segment be x_p (where x_a and x_p belong to the same cover group), and a negative segment from a different song be x_n.

- **Standard Triplet Loss:** The baseline objective forces the distance between the anchor and the positive to be smaller than the distance to the negative by a margin m (empirically set to $m = 0.3$):

$$\mathcal{L}_{Triplet} = \sum_{a,p,n} \max\left(0, \|z_a - z_p\|_2^2 - \|z_a - z_n\|_2^2 + m\right) \tag{1}$$

- **Modified Triplet Hard Loss:** Standard triplet loss quickly produces zero gradients as easy negatives are pushed beyond the margin. To accelerate convergence, we implement a modified Triplet Hard Negative Mining approach. While standard hard mining penalizes both the hardest positive and hardest negative, our implementation uses a random positive but explicitly mines the hardest negative (the closest incorrect track) for each anchor in the batch. This prevents the model from collapsing under the extreme penalty of the furthest positive, while still focusing gradient updates on difficult boundaries:

$$\mathcal{L}_{Hard} = \frac{1}{B}\sum_{i=1}^{B} \max\left(0, \|z_a^{(i)} - z_p\|_2^2 - \min_{x_n}\|z_a^{(i)} - z_n\|_2^2 + m\right) \tag{2}$$

- **Normalized Temperature-scaled Cross Entropy (NT-Xent):** As an alternative contrastive approach, we apply the InfoNCE loss heavily utilized in self-supervised learning frameworks. Rather than comparing an anchor to a single negative, NT-Xent maximizes the agreement between x_a and x_p relative to all other negative segments in the batch simultaneously, scaled by a temperature hyperparameter τ (set to $\tau = 0.1$):

$$\mathcal{L}_{NT-Xent} = -\log \frac{\exp(\text{sim}(z_a, z_p)/\tau)}{\sum_{n=1}^{2N} \mathbb{1}_{[n\neq a]} \exp(\text{sim}(z_a, z_n)/\tau)} \tag{3}$$

  where $\text{sim}(u, v) = u^T v$ is the cosine similarity.

- **Proxy-Anchor Loss:** While Triplet and NT-Xent are instance-based methods that compare pairs or batches of individual segments, we also evaluate Proxy-Anchor loss [5]. This method assigns a learnable "proxy" embedding to each song group. It pulls anchor segments toward their respective class proxy and pushes them away from all other proxies. We utilize this to evaluate whether global space regularization and explicit class centers provide better cluster stability than pure instance-to-instance contrastive learning.

3.9 Retrieval Alignment and Fusion Strategies

Because our model is trained to embed individual temporal segments rather than entire audio files, we must establish strategies for track-level retrieval during evaluation. We implemented and compared six distinct evaluation techniques to bridge the gap between segment-level embeddings and track-level identification:

- **Segment-Level Retrieval:** The baseline approach, where every segment in the query track is compared independently against all segments in the database using cosine similarity.
- **Average Track Pooling (Early Fusion):** Before computing similarities, all segment embeddings belonging to a single track are averaged into one global track-level vector. This drastically speeds up the retrieval process by reducing the search space to a 1-to-1 track comparison.
- **Majority Voting (Late Fusion):** Each segment of the query track independently "votes" for its nearest neighbor in the database. The track that receives the highest number of segment-level votes is selected as the predicted cover. This method provides robustness against isolated segments that may have been poorly embedded due to extreme local noise.
- **Dynamic Time Warping (DTW):** To account for structural misalignments between covers, we compute a Cross-Similarity Matrix (CSM) between all ordered segments of the query track and the database track. We then apply DTW [6] to find the optimal monotonic alignment path through the matrix. While computationally expensive, this method explicitly models the sequential flow of the music.
- **CSM Diagonal Matching:** Rather than computing an expensive full DTW path, this strategy assumes a linear structural correlation and sums the cosine similarities along the main diagonal (and adjacent offset diagonals) of the Cross-Similarity Matrix. This provides a computationally efficient middle ground between unordered voting and full sequence alignment.
- **CSM+CNN Matcher:** A trainable approach where the CSM is treated as a 2D single-channel image and passed through a lightweight Convolutional Neural Network (CNN). The CNN is optimized to recognize diagonal bands and structural patterns that indicate a true cover song match, learning complex alignment heuristics directly from the similarity matrices.

4 Experimental Setup

This section records *what we measure on* and *how experiments were run*. Architectural choices and ablation components are defined in Section 3; per-table baselines are stated in Section 5.

4.1 Dataset and Group-Aware Split

We use a custom cover-song corpus of 120 song groups (240 tracks: one original and one cover per group), sourced from YouTube and Spotify across diverse genres. Tracks are paired by musical identity, not by random sampling.

Group-aware split. Train and validation sets are partitioned by `group_id`, not by individual tracks. With a validation fraction of 0.2, 96 groups (192 tracks) train and 24 groups (48 tracks) validate. This prevents leakage where the original of a song trains while its cover validates (or vice versa). This split concerns **which songs appear in each set**; it is unrelated to the *GroupPairBatchSampler* used during training (Section 3.7).

4.2 Evaluation Metrics

Mean Reciprocal Rank (MRR): Primary retrieval metric, the mean of $1/\text{rank}$ for the first correct cover match. Higher is better.

Top-K accuracy: Fraction of queries where the correct cover ranks in the top K candidates ($K \in \{1, 5\}$).

Cosine Silhouette score: Cluster-quality diagnostic on validation embeddings, measures intra-group cohesion vs. inter-group separation. Reported alongside MRR because ranking alone can mask a poorly structured latent space.

4.3 Training Protocol and Hyperparameters

All runs use cloud NVIDIA GPUs on Kaggle. Shared hyperparameters unless a table states otherwise:

- **Optimizer:** AdamW, learning rate 10^{-4}, weight decay 10^{-4}.
- **LayerPooler (Learned Mix only):** elevated learning rate 10^{-2} (Section 3).
- **Batch size:** 32; mini-batches use either shuffle loading or the *GroupPairBatchSampler* (Section 3.7).
- **Schedule:** up to 50 epochs with early stopping (patience 12 on validation loss); random seed 42.
- **Preprocessing defaults:** 5.0 s segments, 24 kHz mono, dynamic 10/40 segment pool in final configurations (Section 3).

Ablation protocol. Experiments are run as an ordered sequence of studies (backbone, sampling, pooling, retrieval level, segment pool, layers, loss/BN/sampler). Each step locks one configuration that is reused in later comparisons; every results table states the baseline used within its own block.

5 Experimental Results and Analysis

Unless noted otherwise, all experiments follow the shared protocol in Section 4. Within each table, only the component under study changes; the remaining pipeline settings are held fixed to that table's baseline. The commentary below interprets each ablation and does not repeat the full configuration each time.

5.1 Backbone Scale and Hierarchical Layer Ablation

Table 1: Backbone scale and layer aggregation (track-DTW MRR).

Variant	MRR (↑)	Δ MRR	Top-1	Top-5	Silhouette
Backbone scale					
MERT-95M	**0.383**	-	**0.292**	0.417	**-0.080**
MERT-Large (330M)	0.316	-0.067	0.167	**0.458**	-0.155
Layer aggregation					
Layers 12 (last)	**0.519**	-	**0.417**	**0.625**	0.029
Layers 1–3	0.474	-0.045	0.375	0.542	0.015
Layers 4–6	0.483	-0.036	0.375	0.583	0.008
Layers 7–9	0.458	-0.061	0.333	0.583	0.016
Layers 8–10	0.442	-0.077	0.333	0.542	0.023
Layers 10–12	0.445	-0.074	0.292	**0.625**	-0.019
Layers 4–9	0.442	-0.077	0.292	**0.625**	0.011
Mean all layers	0.365	-0.154	0.250	0.417	**0.032**
Learned mix	0.416	-0.103	0.250	0.583	-0.003

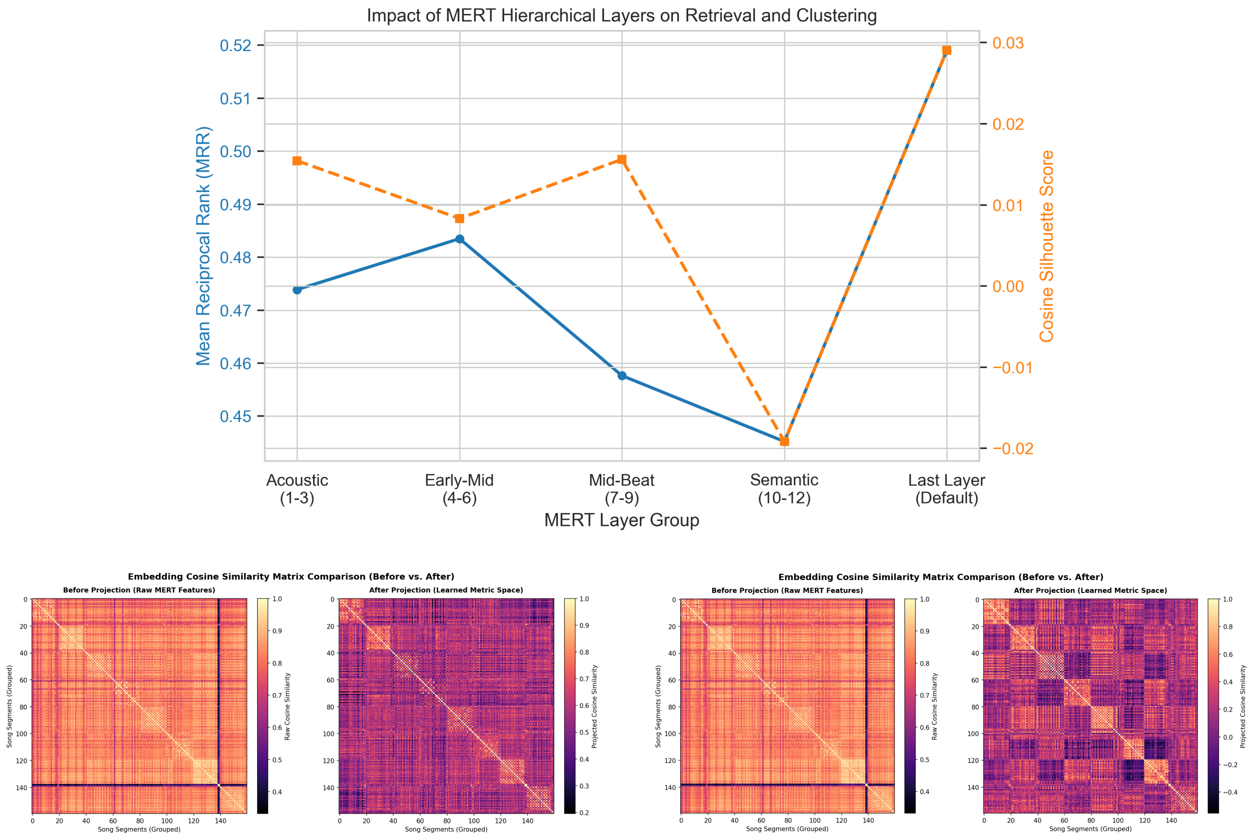
Scaling to MERT-Large (330M) regresses MRR by 0.067 vs. MERT-95M in the backbone-comparison block of Table 1. With only 120 song groups and a frozen encoder, the extra capacity likely amplifies acoustic detail that distinguishes studio recordings rather than musical identity; without backbone fine-tuning, a smaller model is easier to steer through the projection head alone. MERT-Large still improves Top-5 (0.458 vs. 0.417), suggesting that its richer features occasionally surface the correct cover within a broader shortlist even when top-1 ranking suffers.

Among layer subsets, Layers 12 reaches 0.519 MRR; mean-pooling all layers drops MRR by 0.154. Upper MERT blocks encode the most abstract semantics learned during self-supervised pre-training, which aligns well with CSI after metric-learning adaptation. Averaging every layer mixes low-level timbre (layers 1–3) with semantic content, re-introducing exactly the surface-level variance CSI must suppress. The learned softmax mix underperforms fixed subsets (0.416 MRR), plausibly because 120 groups provide limited evidence to tune 13 layer weights reliably, even with an elevated learning rate. Early-mid layers (4–6) remain competitive (0.483 MRR) and retain clearer cross-similarity structure (Figure 3): they capture harmonic and rhythmic cues before the final layer fully commits to higher-level semantics, which can yield stabler segment-to-segment correspondence across covers.

5.2 Segmentation, Pooling, and Dynamic Feeding Strategies

Beat-aligned and stratified sampling each gain ∼0.09 MRR over random cropping. Random 5 s crops often slice through different song sections on the original and the cover; even valid pairs then look like negatives at segment level. Stratified sampling forces coverage of the full timeline, while beat alignment snaps crops to musically meaningful boundaries and partially absorbs tempo drift between versions. Mixed-length segments hurt (0.323 MRR): variable duration adds variance without helping the fixed-length MERT frontend, making embeddings less comparable across tracks.

Max pooling hurts harmonic context (-0.158 vs. mean pool) because it keeps only the strongest frame activations, favoring brief salient events (drum hits, vocal onsets) over the continuous harmonic motion that defines a song identity. Mean pooling aggregates evidence across the whole segment and better matches how covers preserve melody and chord progression.



Layers 12 Layers 4–6

Figure 3: Layer ablation summary (top) and cross-similarity matrices for last layer vs. early-mid layers on a cover pair (bottom).

Table 2: Segmentation, pooling, and dynamic feeding (track-DTW MRR).

Variant	MRR (↑)	∆ MRR	Top-1	Top-5	Silhouette
Sampling					
Random	0.383	-	0.292	0.417	-0.080
Stratified	0.475	+0.092	**0.375**	0.583	-0.105
Mixed	0.323	-0.060	0.208	0.417	-0.163
Beat-aligned	**0.478**	+0.095	0.333	**0.625**	**-0.011**
Temporal pooling					
Mean pooling	**0.478**	-	**0.333**	**0.625**	**-0.011**
Max pooling	0.320	-0.158	0.208	0.417	-0.055
Segment pool					
Fixed pool (10)	0.408	-	0.292	0.500	**-0.045**
Dynamic pool (10/40)	**0.475**	+0.067	**0.375**	**0.583**	-0.105

Dynamic 10/40 segment resampling improves over a fixed 10-segment pool (+0.067 MRR) by exposing the head to different temporal views each epoch. A fixed pool encourages memorization of specific crops; resampling from a larger candidate set acts as cheap temporal augmentation and reduces sensitivity to which 10 zones happened to be pre-selected.

5.3 Batch Normalization and Regularization

We report the effect of batch normalization, pipeline components, and augmentation in Tables 3–5. The three tables answer different questions and are not directly cross-comparable row-for-row.

5.3.1 Batch Normalization across loss functions

Table 3: Effect of Batch Normalization on contrastive losses (track-DTW MRR).

Loss function	MRR (no BN)	MRR (+ BN)	Silhouette (no BN)	Silhouette (+ BN)
Triplet	0.402	0.422	−0.045	0.019
Triplet Hard	**0.538**	0.519	−0.014	0.029
NT-Xent	0.409	0.491	−0.047	0.021
Proxy-Anchor	—	0.460	—	**0.048**

Without BN, Triplet Hard achieves the highest ranking accuracy (MRR 0.538) but negative Silhouette scores indicate a less well-separated embedding space. Hard-negative mining aggressively separates confusing tracks in rank space, yet without normalization the optimizer can stretch clusters into configurations that win retrieval while remaining geometrically fragile. Adding BN trades a small MRR drop for substantially healthier clusters (Silhouette up to +0.029 for Triplet Hard, +0.048 for Proxy-Anchor) by stabilizing hidden activations during projection. NT-Xent benefits more consistently from BN (+0.082 MRR) because batch-wide contrastive denominators amplify scale instability when activations are unnormalized. Proxy-Anchor was only evaluated with BN enabled; its learnable class proxies require a well-scaled space to converge and achieve the best Silhouette in our study.

5.3.2 Sequential ablation (Triplet Hard loss)

Table 4: Sequential ablation of the Triplet Hard pipeline.

Configuration	MRR	ΔMRR	Top-1	Top-5	Silhouette
Baseline	0.4570	—	33.3%	54.2%	0.0043
Group-pair sampler	0.5377	+17.7%	45.8%	58.3%	−0.0144
Group-pair sampler, BN	0.5190	+13.6%	41.7%	62.5%	0.0290
Group-pair sampler, BN, feature aug.	0.4948	+8.3%	41.7%	54.2%	0.0310

The group-pair batch sampler yields the largest MRR gain (+17.7% vs. baseline). Under shuffle batching, most mini-batches lack a cover counterpart for many anchors, so triplet and contrastive losses frequently update on weak or missing positives; pairing originals and covers per group makes every batch structurally useful for CSI. BatchNorm trades a small drop in absolute MRR for healthier clusters (Silhouette +0.0290) and peak Top-5 recall (62.5%): once batching is fixed, normalization mainly reshapes the latent geometry rather than changing which segments co-occur. Feature-level augmentation further stabilizes the embedding space (Silhouette +0.0310) with a modest MRR cost, consistent with light noise/dropout acting as a regularizer on an already strong pipeline rather than teaching new invariances.

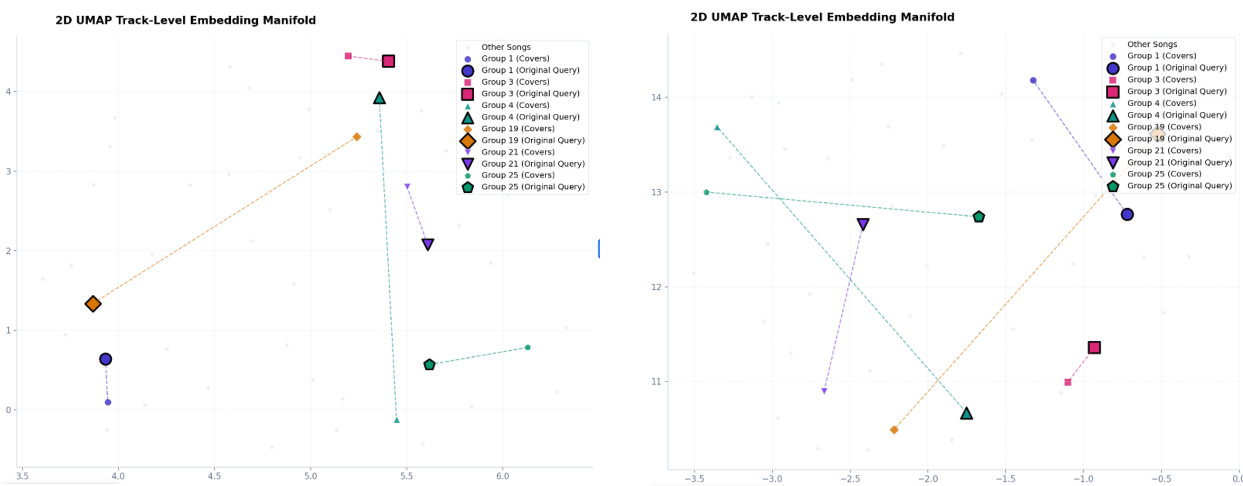
5.3.3 Time-domain vs. feature-level augmentation

Offline waveform augmentation (Setup A) severely degrades retrieval (−21.0% MRR) and collapses the latent space. Aggressive time-stretching in particular can desynchronize aligned crops and blur segment-level correspondence between originals and covers. Feature-level noise and dropout (Setup B) trade a small ranking drop (−4.7% MRR) for the best Silhouette in this block (+0.0310), acting as light regularization on the projection head without altering cached features.

Figure 4 and Figure 5 visualize the BN effect for Triplet Hard with group-pair sampling.

Table 5: Waveform vs. feature-level augmentation.

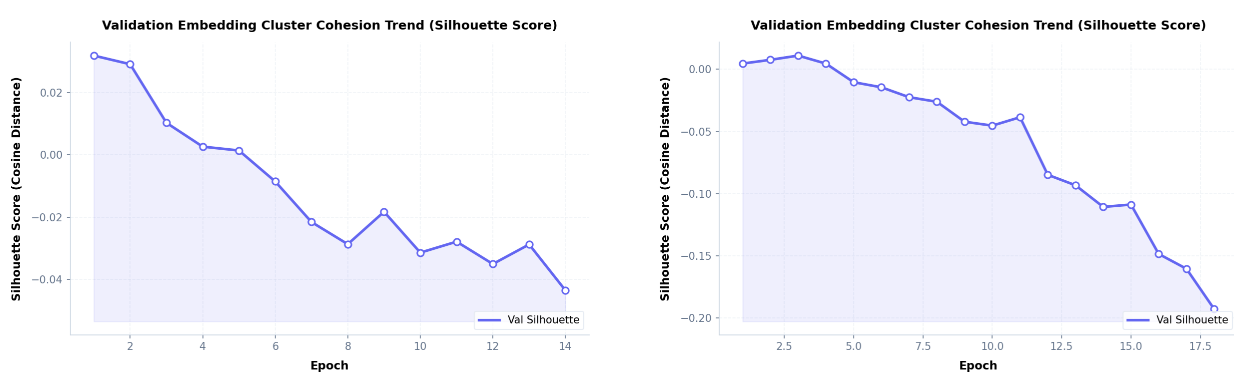
Configuration	MRR	ΔMRR	Top-1	Top-5	Silhouette
Setup A — waveform					
Baseline	0.4570	—	33.3%	54.2%	0.0043
Waveform augmentation	0.3608	−21.0%	20.8%	54.2%	−0.1446
Setup B — feature-level					
Baseline	0.5190	—	41.7%	62.5%	0.0290
Feature augmentation	0.4948	−4.7%	41.7%	54.2%	0.0310



No BN (Shattered Space) With BN (Healthy Clusters)

Figure 4: UMAP projections demonstrating how Batch Normalization prevents dimensional collapse.

To quantify this visual grouping, Figure 5 provides the sample-level Silhouette plots. The positive shift in the silhouette profiles explicitly confirms that BN forces the intra-cluster distances to be significantly tighter than the inter-cluster distances.



No BN (Negative Silhouette) With BN (Positive Silhouette)

Figure 5: Silhouette plots corresponding to the embedding spaces, highlighting the stark mathematical improvement in cluster definition when applying feature-space regularization.

5.4 Loss Formulations and Embedding Space Stability

The loss comparison is reported in Table 3 (Section 5.3). Triplet Hard reaches the best track-DTW MRR (0.538 without BN, 0.519 with BN) because hard-negative mining focuses gradients on the most confusing impostors, the regime that matters for retrieval when covers are timbrally distant. Standard triplet and NT-Xent spread gradient more uniformly across negatives, which is less targeted for fine-grained cover discrimination. Proxy-Anchor with BN yields the highest Silhouette ($+0.048$) by explicitly maintaining a proxy per song group, encouraging a globally coherent cluster layout even when instance-level ranking is slightly lower.

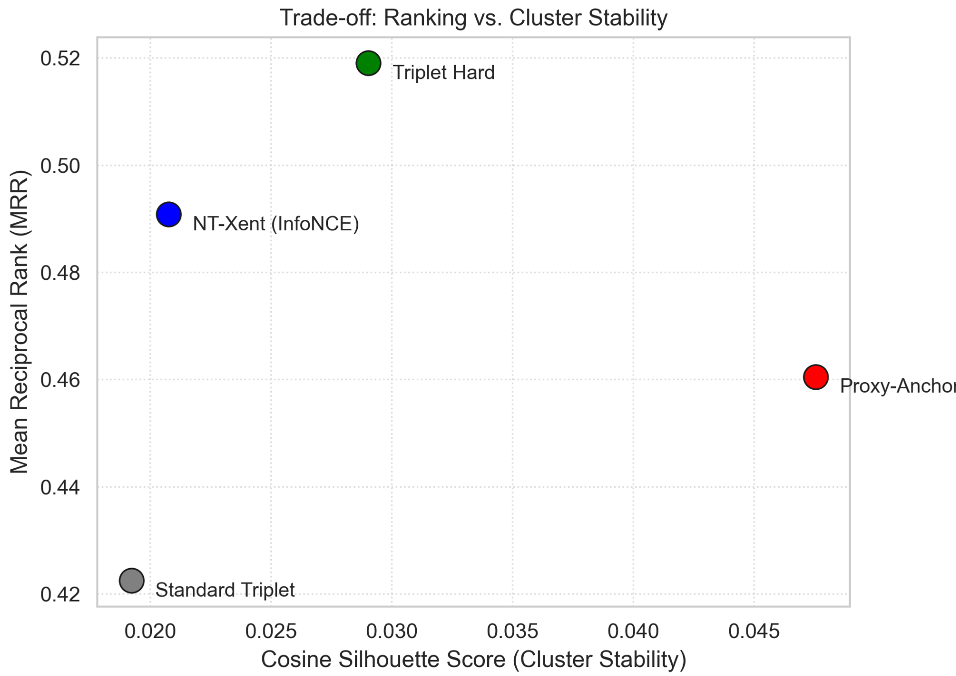


Figure 6: Trade-off between ranking accuracy (MRR) and cluster stability (Silhouette) for different contrastive objectives.

5.5 Information Bottlenecks: The Chroma Capacity Study

We evaluated the effect of explicitly bottlenecking the latent space dimensionality to mirror traditional Chromagram capacities (12 to 96 dimensions).

Table 6: Information Bottleneck Ablation.

Variant	MRR (↑)	Δ MRR	Top-1	Top-5	Silhouette
128-D	**0.519**	-	**0.417**	**0.625**	**0.029**
Chroma-96	0.356	-0.163	0.250	0.458	-0.070
Chroma-48	0.376	-0.143	0.292	0.375	-0.119
Chroma-24	0.280	-0.239	0.167	0.375	-0.186

Contrary to traditional signal-processing assumptions where 24 dimensions perfectly encode pitch classes, forcing the rich hierarchical features of MERT into a 24-dimensional space causes severe collapse (-0.239 MRR). MERT frames already entangle harmony with timbre, dynamics, and performance context; a chroma-sized bottleneck cannot preserve the extra degrees of freedom that separate two renditions of the same song. Performance degrades monotonically as dimensionality shrinks (96-D to 24-D), indicating that CSI here benefits from a moderately high-dimensional semantic space rather than an explicit pitch-class codec.

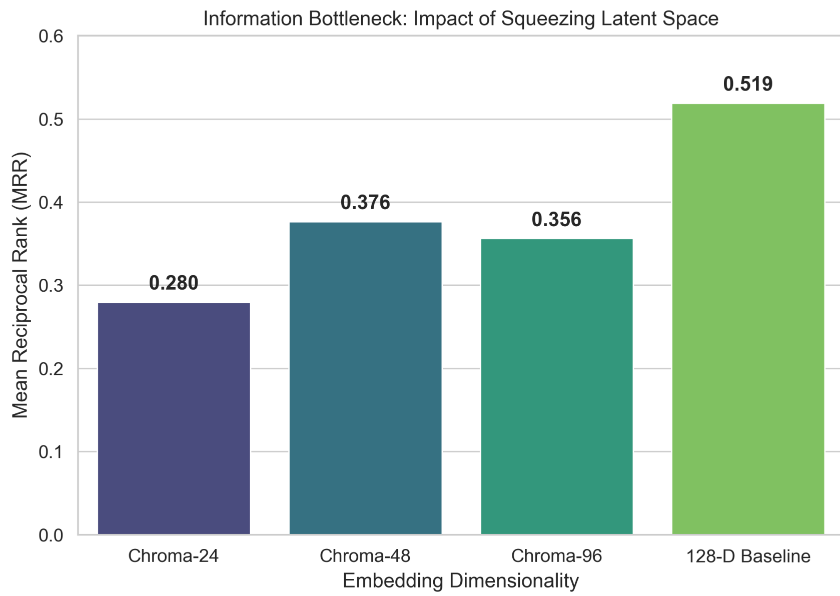


Figure 7: Severe performance regression when squeezing the latent space dimensionality to mirror traditional chromagrams.

Table 7: Track-level retrieval fusion strategies (track-DTW MRR).

Strategy	**MRR** (↑)	**Top-5**
Segment-level	0.265	0.367
Track pooling	0.362	0.583
Majority voting	0.470	0.583
CSM diagonal	0.513	**0.625**
CSM+CNN	0.126	0.167
Track DTW	**0.519**	**0.625**

5.6 Late Fusion Retrieval and Modular CSM Matcher

Segment-level matching is weakest (0.265 MRR) because any single 5 s crop can be atypical or noisy; retrieval then behaves like a fragment search. Track pooling and majority voting aggregate evidence across segments and recover much of the lost signal (0.362–0.470 MRR) by reducing variance from outlier crops. DTW (0.519) and CSM diagonal matching (0.513) perform best because covers often preserve relative segment order even when global tempo shifts; aligning cross-similarity structure along a path captures sequential melody and harmony better than unordered pooling. The learned CSM+CNN matcher (0.126 MRR) underperforms the diagonal heuristic: with only 96 training groups, a second-stage matcher has too little data to learn robust alignment patterns and appears to overfit spurious CSM textures instead of true diagonal structure.

5.7 Optimal System Configurations and Trade-offs

Table 8: Best configuration per evaluation objective.

Objective	**Configuration**	**Score**
Best MRR	Triplet Hard, no BN	**0.538**
Best Top-5	Triplet Hard, with BN	**0.625**
Best Silhouette	Proxy-Anchor	**+0.048**
Best efficiency	Layers 4–6, track pooling	**0.510**

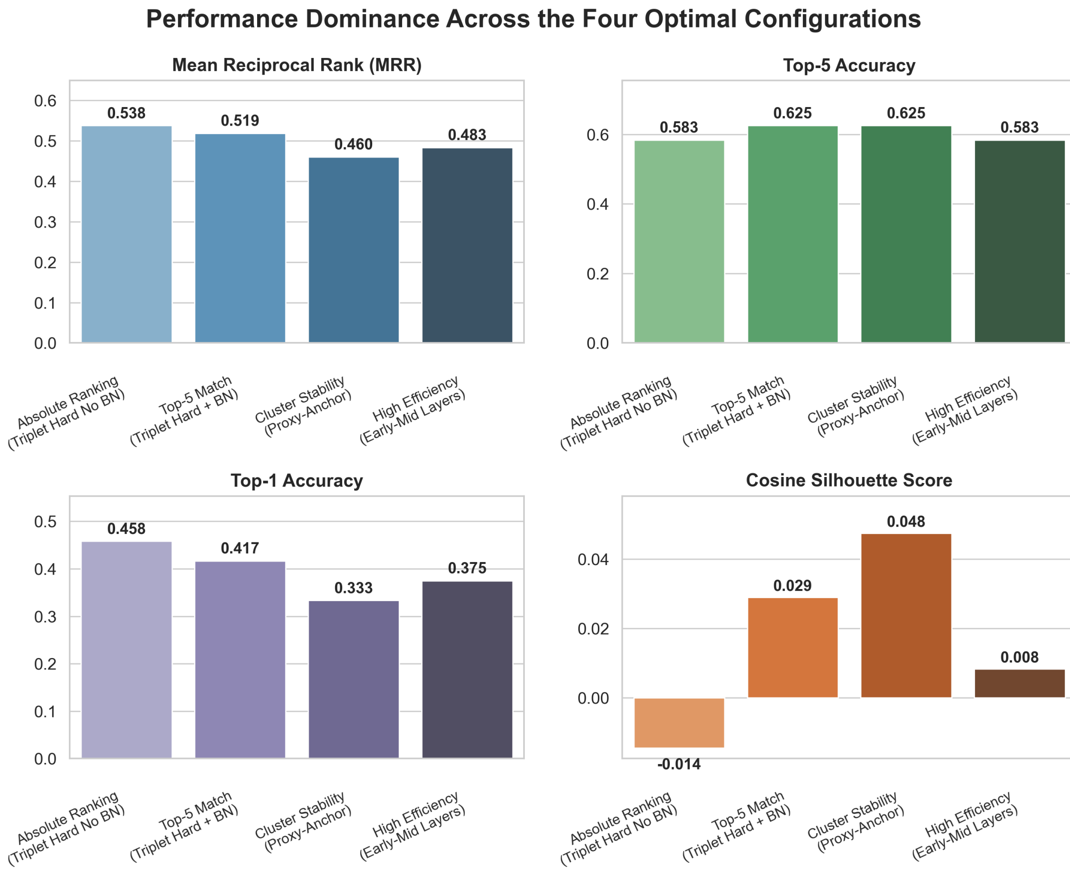


Figure 8: Grouped bar chart comparing the top-performing configurations across the four key evaluation metrics. Each model clearly dominates its optimized objective.

Early-mid layers with mean track pooling reach 0.510 MRR, within $\sim$2% of DTW (0.519) at much lower alignment cost. This supports a practical deployment choice: when latency or compute budget forbids DTW, averaging early-mid segment embeddings at track level is a strong approximation of sequence-aware retrieval. No single row in Table 8 dominates every metric because ranking, Top-5 recall, cluster geometry, and inference cost pull the pipeline in different directions; the table summarizes which configuration to prefer under each objective rather than declaring one universal winner.

5.8 Qualitative Error Analysis

We manually reviewed retrieval outcomes on the 24 validation covers for our best Triplet Hard configuration. Figure 9 groups results into four outcome types; the discussion below focuses on the three failure modes.

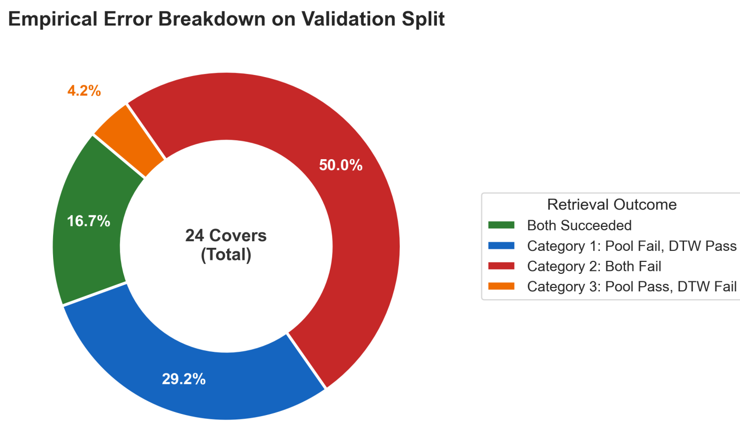


Figure 9: Retrieval outcome breakdown on the validation split (24 covers).

Pool fail, DTW pass (∼29% of errors). Mean track pooling often fails when a cover changes arrangement or instrumentation enough to blur the averaged embedding, e.g., vocal-pop or acoustic reinterpretations of rock originals, yet DTW still retrieves the true original by aligning segment sequences chronologically. Static pooling loses structural identity under heavy timbral noise; sequence alignment preserves phrase order and recovers the match.

Both fail (∼50% of errors). The largest share of errors affects both pooling and DTW. These are extreme reinterpretations that alter harmony, rhythm, or production so much that frozen MERT features for cover and original diverge beyond what a shallow projection head can bridge, folk-rock electrifications, funk reworkings of pop tracks, or acoustic strips of industrial originals. They mark the practical ceiling of cached-embedding retrieval without backbone fine-tuning or stronger pitch-invariant training.

Pool pass, DTW fail (∼4% of errors). Less often, mean pooling succeeds while DTW fails. A typical case is a cover that keeps key and harmony but inserts non-monotonic structur, such as a fade-out followed by a re-entry with altered tempo, which penalizes the DTW path and can pull alignment toward an unrelated linear track. This shows DTW's sensitivity to structural edits, not just timbral change.

Overall, temporal alignment helps when timbre shifts but phrase order is preserved; neither fusion strategy handles the hardest stylistic gaps under a frozen backbone; and monotonic aligners remain vulnerable to deliberate arrangement breaks.

6 Conclusion and Future Work

In this work, we conducted a comprehensive ablation study comprising over 36 distinct experiments to identify strong configurations for Cover Song Identification (CSI). Triplet Hard with group-pair sampling reaches 0.538 track-DTW MRR (without BN); with BN, retrieval stays competitive (0.519) while cluster quality improves. Early-mid layer features with mean track pooling achieve 0.510 MRR, close to DTW (0.519) without sequence alignment. Batch Normalization and Proxy-Anchor loss mainly improve embedding-space structure (Silhouette up to $+0.048$).

While our established configurations achieve robust efficiency and cluster stability, there are several promising, accessible avenues for future research. First, **Energy Filtering** could be applied prior to the temporal pooling step; by dynamically dropping silent or low-energy frames, we can prevent non-musical segments from diluting the final mean-pooled embedding. Second, we propose exploring **Circular Chroma Rolling** as an online data augmentation technique, effectively simulating key transpositions during training to force the network to learn key-invariant representations. Finally, integrating a **Supervised Pitch Multi-Task** objective, where the projection head is explicitly trained to predict the underlying pitch class alongside the contrastive loss, could further disentangle melodic semantics from acoustic noise without requiring massive increases in parameter scale.

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A Custom Dataset Properties and Catalog

To verify the deliverables of this research and facilitate complete reproducibility, this appendix provides a comprehensive breakdown of the custom Cover Song Identification (CSI) dataset. The dataset consists of 120 hand-curated song groups, totaling 240 unique tracks.

Table 9 details the acoustic and stylistic distribution of the dataset across primary musical genres, tempo shifts, and vocal-to-instrumental transformations. Quantifying this variance ensures that the model is evaluated against a realistic and highly challenging acoustic domain gap.

Table 9: Custom Dataset Distribution Metrics (120 Song Groups, 240 Tracks).

Dimension & Category	**Song Groups**	**Percentage (%)**
Primary Musical Genre		
Rock / Alternative / Metal	52	43.3%
Pop / Electronic / Synth-Pop	38	31.7%
Folk / Country / Blues	18	15.0%
Soul / R&B / Jazz / Funk	12	10.0%
Tempo Shifts (Original $\rightarrow$ Cover)		
Similar Tempo ($\pm$15 BPM)	60	50.0%
Fast-to-Slow	35	29.2%
Slow-to-Fast	25	20.8%
Vocal vs. Instrumental Shifts		
Vocal-to-Vocal	108	90.0%
Vocal-to-Instrumental	8	6.7%
Instrumental-to-Vocal	4	3.3%

Following the statistical distribution, Table 10 provides the complete catalog of all evaluated track pairs, listing the reference original recording alongside the specific cover version used for retrieval alignment.

Table 10: Custom Dataset Song Groups and Track Pairs.

ID	**Reference Original (Artist – Title)**	**Evaluated Cover (Artist – Title)**
1	Nine Inch Nails – *Hurt*	Johnny Cash – *Hurt*
2	Leonard Cohen – *Hallelujah*	Jeff Buckley – *Hallelujah*
3	Bob Dylan – *All Along the Watchtower*	Jimi Hendrix – *All Along the Watchtower*
4	Tears For Fears – *Mad World*	Gary Jules, Michael Andrews – *Mad World*
5	Dolly Parton – *I Will Always Love You*	Whitney Houston – *I Will Always Love You*

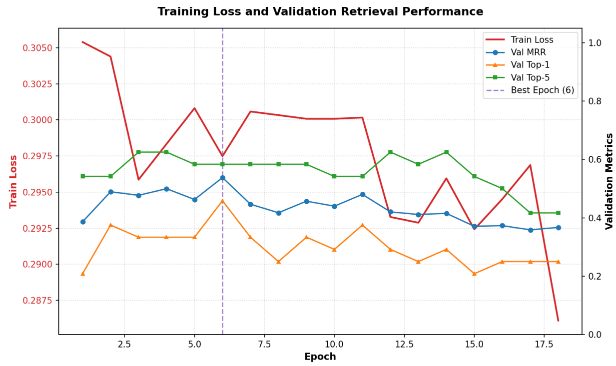
ID	Reference Original (Artist – Title)	Evaluated Cover (Artist – Title)
6	The Top Notes – *Twist And Shout*	The Beatles – *Twist And Shout - Remastered 2009*
7	Otis Redding – *Respect - Mono; 2008 Remaster*	Aretha Franklin – *Respect*
8	Prince – *Nothing Compares 2 U*	Sinéad O'Connor – *Nothing Compares 2 U*
9	Gloria Jones – *Tainted Love*	Soft Cell – *Tainted Love*
10	Michael Jackson – *Smooth Criminal - 2012 Remaster*	Alien Ant Farm – *Smooth Criminal*
11	Bob Dylan – *Knockin' On Heaven's Door*	Guns N' Roses – *Knockin' On Heaven's Door*
12	Roberta Flack – *Killing Me Softly With His Song*	Fugees, Ms. Lauryn Hill – *Killing Me Softly With His Song*
13	Big Mama Thornton – *Hound Dog*	Elvis Presley – *Hound Dog*
14	Robert Hazard, (lee junglim) – *Girls Just Want TO Have Fun (Grand Allegro 1)*	Cyndi Lauper – *Girls Just Want to Have Fun*
15	David Bowie – *The Man Who Sold the World - 2015 Remaster*	Nirvana – *The Man Who Sold The World - Live*
16	Simon & Garfunkel – *The Sound of Silence*	Disturbed, CYRIL – *The Sound of Silence - CYRIL Remix*
17	Marvin Gaye – *I Heard It Through The Grapevine*	Creedence Clearwater Revival – *I Heard It Through The Grapevine*
18	The Bobby Fuller Four – *I Fought the Law - Single Version*	The Clash – *I Fought the Law*
19	The Beatles – *Got To Get You Into My Life - Remastered 2009*	Earth, Wind & Fire – *Got to Get You Into My Life*
20	Fleetwood Mac – *Black Magic Woman*	Santana – *Black Magic Woman*
21	The Dubliners – *Whiskey in the Jar*	Metallica – *Whiskey In The Jar*
22	Stevie Wonder – *Higher Ground*	Red Hot Chili Peppers – *Higher Ground - Remastered 2003*
23	James Ray – *I've Got My Mind Set on You*	George Harrison – *Got My Mind Set on You*
24	The Troggs – *Wild Thing*	Jimi Hendrix – *Wild Thing - Live At Monterey*
25	Badfinger – *Without You - Remastered 2010*	Mariah Carey – *Without You*
26	Little Richard – *Tutti Frutti*	Pat Boone – *Tutti Frutti*
27	Sam Cooke – *Wonderful World*	Louis Armstrong – *What A Wonderful World*
28	Ben E. King – *Stand By Me*	John Lennon – *Stand By Me - Remastered 2010*
29	Eurythmics, Annie Lennox, Dave Stewart – *Sweet Dreams (Are Made of This) - 2005 Remaster*	Marilyn Manson – *Sweet Dreams (Are Made Of This)*
30	The Leaves – *Hey Joe*	Jimi Hendrix – *Hey Joe*
31	Brenda Lee – *Always On My Mind*	Willie Nelson – *Always On My Mind*
32	Carl Perkins – *Blue Suede Shoes*	Elvis Presley – *Blue Suede Shoes*
33	Bing Crosby – *Try A Little Tenderness*	Otis Redding – *Try a Little Tenderness*
34	Anthony Newley, Leslie Bricusse, Downshire Brass, Brendan Coyle – *Feeling Good*	Nina Simone – *Feeling Good*
35	Bob Dylan – *Make You Feel My Love*	Adele – *Make You Feel My Love*
36	TOTO – *Africa*	Weezer – *Africa*
37	a-ha – *Take on Me*	Weezer – *Take on Me*
38	Wham! – *Last Christmas*	Taylor Swift – *Last Christmas*
39	Dolly Parton – *Jolene*	Miley Cyrus – *Jolene - Live*
40	The Knife – *Heartbeats*	José González – *Heartbeats*
41	DNA, Suzanne Vega, Neal Slateford, Nick Batt – *Tom's Diner - 7" Version*	DNA, Suzanne Vega, Neal Slateford, Nick Batt – *Tom's Diner - 7" Version*
42	Britney Spears – *Toxic*	Yael Naim – *Toxic*
43	Nirvana – *Smells Like Teen Spirit*	Tori Amos – *Smells Like Teen Spirit - 2015 Remaster*

ID	Reference Original (Artist – Title)	Evaluated Cover (Artist – Title)
44	Chris Isaak – *Wicked Game*	James Vincent McMorrow – *Wicked Game - Recorded Live at St Canice Cathedral, Kilkenny*
45	Depeche Mode – *Personal Jesus*	Johnny Cash – *Personal Jesus*
46	Soundgarden – *Black Hole Sun*	Norah Jones – *Black Hole Sun - Live*
47	Simon & Garfunkel – *Bridge Over Troubled Water*	Aretha Franklin – *Bridge over Troubled Water*
48	Derek & The Dominos – *Layla*	Benne – *Layla*
49	Neil Young – *Helpless*	k.d. lang – *Helpless*
50	Joni Mitchell – *Big Yellow Taxi*	Counting Crows, Vanessa Carlton – *Big Yellow Taxi*
51	New Order – *Bizarre Love Triangle*	Frente! – *Bizarre Love Triangle*
52	Don Henley – *The Boys Of Summer - Remastered 2024*	The Ataris – *The Boys of Summer*
53	Queen – *Crazy Little Thing Called Love - Remastered 2011*	Michael Bublé – *Crazy Little Thing Called Love*
54	Commodores – *Easy*	Faith No More – *Easy*
55	Tracy Chapman – *Fast Car*	Jonas Blue, Dakota – *Fast Car*
56	Tom Petty – *Free Fallin'*	John Mayer – *Free Fallin' - Live at the Nokia Theatre, Los Angeles, CA - December 2007*
57	Creedence Clearwater Revival – *Have You Ever Seen The Rain*	Rod Stewart – *Have You Ever Seen The Rain*
58	The Beatles – *Hey Jude*	Wilson Pickett – *Hey Jude*
59	B.J. Thomas – *Hooked On A Feeling*	Blue Swede, Björn Skifs – *Hooked On A Feeling*
60	Gloria Gaynor – *I Will Survive*	CAKE – *I Will Survive*
61	Genesis – *Land of Confusion - 2007 Remaster*	Disturbed – *Land of Confusion*
62	The Beatles – *Let It Be - Remastered 2009*	Aretha Franklin – *Let It Be*
63	The Doors – *Light My Fire*	José Feliciano – *Light My Fire*
64	Bob Dylan – *Like a Rolling Stone*	Jimi Hendrix – *Like A Rolling Stone - Live At Monterey*
65	Little Eva – *The Loco-Motion*	Kylie Minogue – *The Loco-Motion*
66	The Everly Brothers – *Love Hurts - 2007 Remaster*	Nazareth – *Love Hurts*
67	The Beatles – *Lucy In The Sky With Diamonds - Remastered 2009*	Elton John – *Lucy In The Sky With Diamonds*
68	Van Morrison – *Moondance - 2013 Remaster*	Michael Bublé – *Moondance*
69	Simon & Garfunkel – *Mrs. Robinson - From "The Graduate" Soundtrack*	The Lemonheads – *Mrs. Robinson - 2022 Remastered Edition*
70	Frank Sinatra – *My Way*	Sid Vicious – *My Way*
71	Bob Marley & The Wailers – *No Woman No Cry*	Fugees – *No Woman No Cry*
72	The Rolling Stones – *Paint It Black - Live / Remastered 2009*	Vanessa Carlton – *Paint It Black*
73	The Hit Crew – *Oh Pretty Woman (As Made Famous By: Roy Orbison)*	Van Halen – *(Oh) Pretty Woman - 2015 Remaster*
74	Creedence Clearwater Revival – *Proud Mary*	Ike & Tina Turner – *Proud Mary*
75	Bob Marley & The Wailers – *Redemption Song*	Johnny Cash, Joe Strummer – *Redemption Song*
76	Elton John – *Rocket Man (I Think It's Going To Be A Long, Long Time)*	Kate Bush – *Rocket Man*
77	The White Stripes – *Seven Nation Army*	Ben L'Oncle Soul – *Seven Nation Army - Remasterisée*
78	Taylor Swift – *Shake It Off*	Ryan Adams – *Shake It Off*
79	Billy Joel – *She's Always a Woman*	Fyfe Dangerfield – *She's Always A Woman - Bonus Track*
80	The Platters – *Smoke Gets In Your Eyes*	Bryan Ferry – *Smoke Gets In Your Eyes*

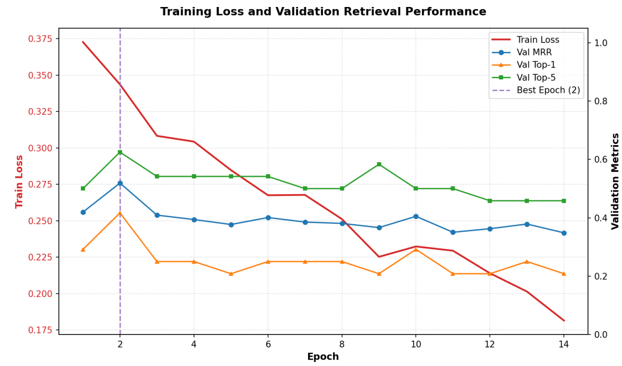
ID	Reference Original (Artist – Title)	Evaluated Cover (Artist – Title)
81	MGM Orchestra & Judy Garland – *Finale - Somewhere Over The Rainbow - (Digitally Remastered 2010)*	Israel Kamakawiwo'ole – *Somewhere Over The Rainbow/What A Wonderful World*
82	The Beatles – *Strawberry Fields Forever - Remastered 2009*	The Beatles – *Strawberry Fields Forever - Remastered 2009*
83	Mark James – *Suspicious Minds*	Elvis Presley – *Suspicious Minds*
84	Guns N' Roses – *Sweet Child O' Mine*	Sheryl Crow – *Sweet Child O' Mine*
85	Al Green – *Take Me to the River*	Talking Heads – *Take Me to the River*
86	Cyndi Lauper – *Time After Time*	Eva Cassidy – *Time After Time (Time After Time)*
87	Rihanna, JAŸ-Z – *Umbrella*	Train – *Umbrella*
88	Queen, David Bowie – *Under Pressure - Remastered 2011*	My Chemical Romance, The Used – *Under Pressure*
89	The Zutons – *Valerie*	Mark Ronson, Amy Winehouse – *Valerie (feat. Amy Winehouse) - Version Revisited*
90	Jimi Hendrix – *Voodoo Child (Slight Return)*	Stevie Ray Vaughan – *Voodoo Child (Slight Return)*
91	Queen – *We Will Rock You - Remastered 2011*	Five, Queen – *We Will Rock You (feat. Queen) - Radio Edit*
92	Lead Belly – *Where Did You Sleep Last Night*	Nirvana – *Where Did You Sleep Last Night - Live*
93	Pink Floyd – *Wish You Were Here*	Wyclef Jean – *Wish You Were Here*
94	The Beatles – *Yesterday - Remastered 2009*	Frank Sinatra – *Yesterday*
95	Billy Preston – *You Are So Beautiful*	Joe Cocker – *You Are So Beautiful*
96	The Cranberries – *Zombie*	Bad Wolves – *Zombie*
97	Radiohead – *Creep*	Scott Bradlee's Postmodern Jukebox, Haley Reinhart – *Creep*
98	Oasis – *Wonderwall - Remastered*	Ryan Adams – *Wonderwall*
99	Aerosmith – *Walk This Way*	Run–D.M.C., Aerosmith – *Walk This Way (feat. Aerosmith)*
100	Bonnie Tyler – *Total Eclipse of the Heart*	Westlife – *Total Eclipse of the Heart*
101	U2 – *With Or Without You*	Keane – *With Or Without You - BBC Jo Whiley Session / 2004*
102	Green Day – *Boulevard of Broken Dreams*	Green Day – *Boulevard of Broken Dreams*
103	Snow Patrol – *Chasing Cars*	Sleeping At Last – *Chasing Cars*
104	Eagles – *Hotel California - 2013 Remaster*	Gipsy Kings – *Hotel California - Spanish Mix*
105	John Lennon – *Imagine - Remastered 2010*	A Perfect Circle – *Imagine*
106	Bob Dylan – *Mr. Tambourine Man*	The Byrds – *Mr. Tambourine Man*
107	Billie Holiday – *Summertime*	Big Brother & The Holding Company, Janis Joplin – *Summertime*
108	Hoagy Carmichael – *Georgia On My Mind*	Ray Charles – *Georgia on My Mind*
109	Julie London – *Cry Me A River*	Michael Bublé – *Cry Me a River*
110	Louis Armstrong – *Mack the Knife*	Bobby Darin – *Mack the Knife*
111	Little Willie John – *Fever*	Peggy Lee – *Fever*
112	Nancy Sinatra – *These Boots Are Made for Walkin'*	Jessica Simpson – *These Boots Are Made for Walkin'*
113	The Dixie Cups – *Iko Iko*	The Belle Stars – *Iko Iko (From "Rain Man")*
114	Bob Dylan – *House Of The Rising Sun - Informal Recording, NYC, 1963*	The Animals – *House of the Rising Sun*
115	Simon & Garfunkel – *A Hazy Shade of Winter*	The Bangles – *Hazy Shade of Winter*
116	Cyndi Lauper – *True Colors*	Phil Collins – *True Colors - 2016 Remaster*
117	The Who – *Behind Blue Eyes*	Limp Bizkit – *Behind Blue Eyes*
118	Toni Basil – *Mickey*	Toni Basil – *Hey Mickey*
119	The Foundations – *Build Me Up Buttercup - Mono*	The Foundations – *Build Me Up Buttercup - Mono*
120	The Rolling Stones – *(I Can't Get No) Satisfaction - Mono*	DEVO – *Satisfaction*

B Training Convergence and Stability

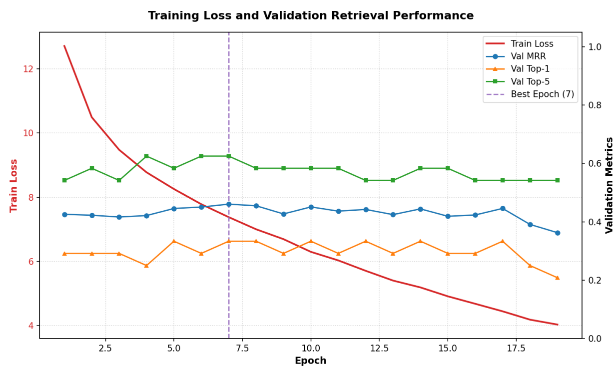
To demonstrate the training stability and convergence speed of the proposed architectures, Figure 10 provides the training and validation loss curves for the four optimal configurations highlighted in Section 5.



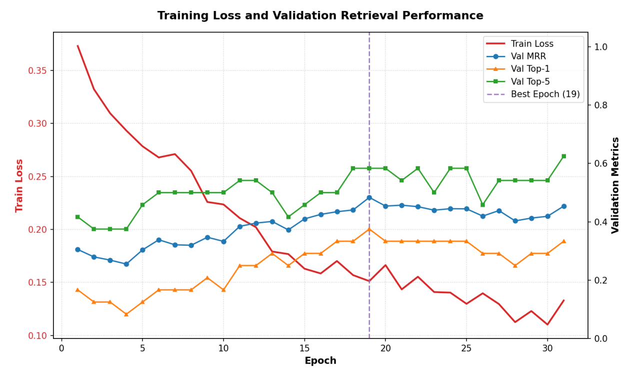
Triplet Hard (Without BN)



Triplet Hard (With BN)



Proxy-Anchor



Early-Mid Layers (Mean Pool)

Figure 10: Training and validation loss curves for the four SOTA models. The inclusion of Batch Normalization noticeably smooths the optimization trajectory.